# SASTRA DEEMED UNIVERSITY

(A University under section 3 of the UGC Act, 1956)

**End Semester Examinations**

**May 2023**

Course Code: **INT405R01**

Course: **MACHINE LEARNING TECHNIQUES**

QP No. :**U288-6**

Duration: **3** hours

Max. Marks:**100**

**PART – A**

**Answer any FOUR questions** **4 x 20 = 80 Marks**

1. a. Only 1 in 1000 adults is afflicted with a rare disease for which a diagnostic test has been developed. The test is such that when an individual actually has the disease, a positive result will occur 99% of the time, whereas an individual without the disease will show a positive test result only 2 % of the time. If a randomly selected individual is tested and the result is positive, what is the probability that the individual has the disease? (10)

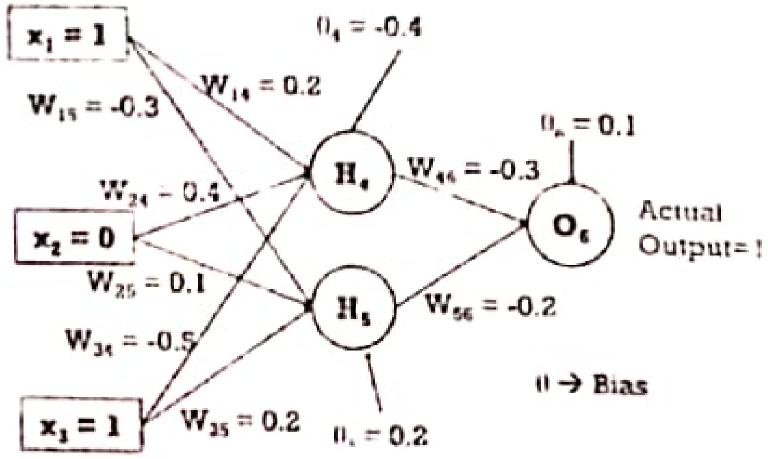
   b. Given are the two variations of blood pressure: systolic BP & diastolic BP. Compute PCA to discover the principal components of the two strongly correlated variables. Use PCA Algorithm to transform the pattern (79, 127) onto the eigen vector. (10)

| Diastolic BP | Systolic BP |
|---|---|
| 78 | 126 |
| 80 | 128 |
| 81 | 127 |
| 82 | 130 |
| 84 | 130 |
| 86 | 132 |

2. a. Explain how Newton-Raphson Iterative Optimization Scheme can applied for minimizing the Cross-Entropy error function E(w) involved in Logistic Regression (IRLS) model? (10)

b. Minimize $f(x_1, x_2) = x_1^2 - 3x_1 x_2 + 2x_2^2$ steepest Descent Scheme. (10)

3. Consider the 3-2-1 architecture and update the weights of the neural network for one iteration using sigmoid activation function. Assume that the actual output is 1 and the learning rate is 0.9 and run the iteration once.

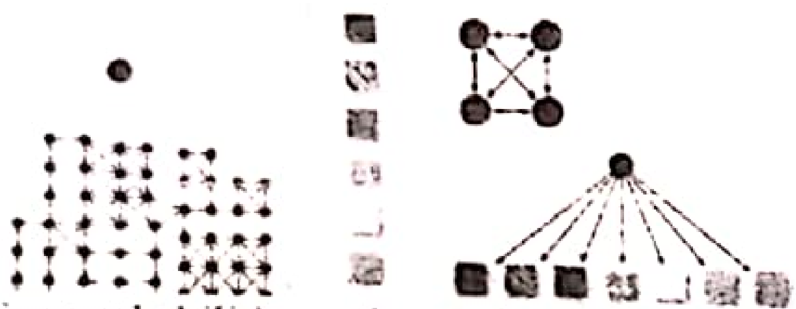



4. a. Write short note on 'Sparse Representations', 'Bagging' and 'Drop out' in Regularization for Deep Learning Schemes. (10)

b. Construct a Hyperplane for classifying the following data points using SVM classifier algorithm also find support vectors.
Class 1: (1,2) and Class 2: (-1,2) and (-1, -1). (10)

5. a. A robot observes various colors of terrains at each of it states(position)using sensors, and based on the color of the terrains, it can localize itself. The state representation is as shown in the figure below.

$$O = \{O_1, O_2, O_3, \ldots, O_T\}$$



The transition probabilities and emission probabilities are known. A sequence of observation data (color of the terrain) is given to you. As a Machine learning expert, you are expected to compute the sequence of states the robot could have travelled for the given observation using Viterbi algorithm. How do you solve this problem? Give appropriate mathematical expressions and reasons. (10)

b. Describe the Fisher's Discriminant Classifier Model with its distinguishing features from Linear Discriminant Analysis by providing an example. (10)

6. It is observed that the initial probabilities for a person to apply Hair oil, Shampoo and Natural products are 0.25, 0.25 and 0.5 respectively. The probabilities for transition from Hairoil to Hairoil, Hairoil to Shampoo, Hair oil to Natural products are 0.5, 0.2 and 0.3 respectively. The probabilities for transition from Shampoo to Hairoil, Shampoo to Shampoo and Shampoo to Natural products are 0.4, 0.2 and 0.4 respectively. And the transition probabilities for transition from Natural products to Hairoil, Natural products to Shampoo and Natural products to Natural products are 0.2, 0.2 and 0.6 respectively. The probability for a person to get short, medium and long hair from Hairoil is 0.3, 0.3, 0.4 respectively. The probability for a person to get short(S), medium(M) and long(L) hair from shampoo is 0.5, 0.2, 0.3 respectively. The probability for a person to get short, medium and long hair from Natural products is 0.1, 0.2, 0.7 respectively. From the above data, evaluate the probability of an observation sequence LSM by using forward algorithm.

## PART -B

**Answer the following** **1 x 20 = 20 Marks**

7. a. X and Y are two random variables having the joint density function, $f(x, y) = (1/27)(x+2y)$, where x and y can assume only the integer values 0,1 and 2, find the marginal distributions and conditional distribution of Y for X = x. (5)

b. Build a Logistic regression classifier for the following data with two input features and targeted output labels also find the class to which the point (7,9) lies with its associated probability. For optimizing the parameters use the initial approximations (-3, 0.6, -1.2) (5)

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| X | 2 | 4 | 5 | 6 | 8 | 9 | 10 |
| Y | 3 | 5 | 7 | 8 | 9 | 11 | 13 |
| Targeted outcome | 0 | 0 | 0 | 1 | 1 | 1 | 1 |

c. A gambler's luck follows a pattern. If he wins a game, the probability of his winning the next game is 0.6. However, if he loses a game, the probability of his losing the next game is 0.7. There is an even chance that the gambler wins first game. What is the probability that he wins?
(i) the second game, (ii) the third game (iii) in the long run. (5)

d. Explain 'convolution and pooling as an infinitely strong prior' in convolutional Networks. (5)

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